

BHISHO, South Africa

WMO: 687520

Lat: 32.894S

Lon: 27.287E

Elev: 590

StdP: 94.44

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 5.2	(c) 6.1	(d) -7.5	(e) 2.1	(f) 18.2	(g) -5.3	(h) 2.6	(i) 18.0	(j) 14.9	(k) 11.8	(l) 13.0	(m) 13.0	(n) 4.7	(o) 260	(p) 0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 9.3	(c) 33.2	(d) 19.6	(e) 31.1	(f) 19.6	(g) 29.3	(h) 19.1	(i) 22.9	(j) 29.0	(k) 22.0	(l) 27.6	(m) 21.2	(n) 26.2	(o) 4.9	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.1	25.1	20.3	16.1	24.0	19.6	15.4	23.2	71.0	29.1	67.4	27.9	64.3	26.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.4	9.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			1.4	25.3	1.1	1.3	0.5	26.2	-0.1	27.0	-0.8	27.8	-1.6	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	20.8	21.2	20.0	18.1	16.5	14.3	14.0	14.8	15.7	16.4	17.6	19.7
	DBStd	4.15	3.07	3.03	3.22	3.49	3.11	3.19	3.54	3.80	3.69	3.66	3.19	3.17
	HDD10.0	13	0	0	0	0	0	3	6	3	2	0	0	0
	HDD18.3	798	11	7	18	46	76	126	141	123	96	82	51	22
	CDD10.0	2719	336	314	310	243	202	132	129	152	172	200	228	301
	CDD18.3	463	87	88	71	39	19	4	6	14	17	23	29	64
	CDH23.3	3464	596	560	468	271	139	43	54	167	197	235	238	497
(14) Wind	WSAvg	4.6	4.8	4.6	4.3	4.1	4.1	4.5	4.5	4.8	4.8	5.0	5.1	4.8
	PrecAvg	728	91	80	87	51	26	26	25	37	47	73	92	94
	PrecMax	906	161	144	237	126	114	161	115	154	206	202	236	226
	PrecMin	524	36	36	14	13	5	2	0	5	8	15	11	20
	PrecStd	95	37	26	44	30	27	33	31	33	45	47	51	51
	DB	34.7	34.6	34.7	32.4	29.2	26.8	26.9	31.3	32.7	34.0	32.8	34.4	
	MCWB	21.6	21.9	20.2	17.7	16.5	13.1	12.2	14.8	15.7	17.2	17.7	20.8	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	31.5	31.2	30.8	29.4	27.0	24.6	24.9	27.4	28.2	29.6	28.7	31.0	
	2% MCWB	21.8	21.7	19.5	17.2	15.5	12.0	11.7	13.0	15.6	17.2	18.9	20.9	
	5% DB	28.8	28.9	28.0	26.5	24.6	22.6	22.7	24.7	25.1	25.9	26.0	28.4	
	5% MCWB	21.1	21.4	19.8	17.3	14.4	11.5	11.1	12.4	14.8	16.6	18.5	20.0	
	10% DB	26.7	26.7	25.6	24.1	22.1	20.3	20.2	22.0	22.2	23.1	23.7	25.9	
	10% MCWB	20.1	20.6	19.1	16.9	13.8	10.6	10.4	11.6	14.1	15.8	17.5	19.3	
	WB	24.0	24.3	22.5	20.7	19.3	15.0	14.5	17.2	19.6	20.5	21.9	23.6	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	30.1	30.2	29.2	25.0	23.9	19.4	18.8	24.0	26.4	27.4	27.6	30.9	
	2% WB	22.7	23.0	21.3	19.6	16.8	13.7	13.5	15.6	17.3	18.8	20.2	21.9	
	2% MCDB	28.9	28.4	27.0	24.9	21.8	19.0	18.7	21.4	23.4	24.8	25.8	28.1	
	5% WB	21.7	22.0	20.5	18.7	15.8	12.9	12.7	14.6	16.1	17.7	19.1	21.0	
	5% MCDB	27.0	27.1	25.7	23.5	20.7	19.0	18.4	19.9	21.6	23.5	24.0	26.2	
	10% WB	20.7	21.1	19.6	17.7	14.9	12.1	11.9	13.6	15.1	16.7	18.1	20.0	
	10% MCDB	25.4	25.5	24.3	22.0	20.2	17.9	17.4	18.1	20.2	21.8	22.2	24.5	
(35) Mean Daily Temperature Range	MDBR	9.5	9.3	9.6	9.8	9.9	10.5	10.3	10.9	11.2	10.1	9.8	9.5	
	5% DB	14.1	13.6	14.2	13.6	13.5	13.6	13.6	15.8	16.7	15.5	14.5	15.1	
	5% MCWBR	5.8	5.5	5.5	5.1	5.6	5.6	5.1	5.5	6.5	6.4	6.3	6.5	
	5% WB	12.4	12.4	12.1	11.5	10.9	11.2	10.6	11.7	13.5	12.1	12.1	12.5	
	5% MCWBR	5.9	5.5	5.5	5.2	5.4	5.3	5.0	5.2	6.4	5.9	6.2	6.1	
(40) Clear-Sky Solar Irradiance	taub	0.380	0.380	0.364	0.335	0.317	0.302	0.306	0.356	0.384	0.393	0.372	0.376	
	taud	2.433	2.441	2.460	2.510	2.531	2.542	2.524	2.376	2.295	2.315	2.412	2.437	
	Ebn at Noon	958	938	916	890	856	846	860	852	880	913	959	966	
	Edh at Noon	122	117	107	91	79	73	78	102	124	131	124	123	
(44) All-Sky Solar Radiation	RadAvg	6.53	5.99	5.08	4.04	3.21	2.87	3.08	3.80	4.83	5.59	6.38	6.71	
	RadStd	0.37	0.33	0.39	0.23	0.13	0.16	0.18	0.24	0.34	0.45	0.39	0.45	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.35	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-85	+142	+45

Nomenclature: See separate page